Conférence invitée – Université Technologique de Compiègne
UR06 “Géomatique” - 15 décembre 2021

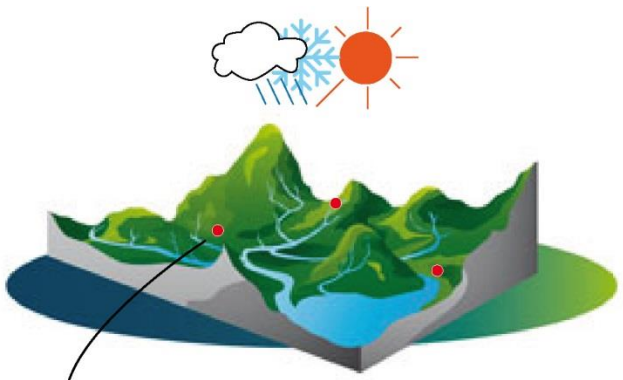


An inverse snow-hydrological modelling approach to infer elevation gradients of air temperature and precipitation in mountainous basins

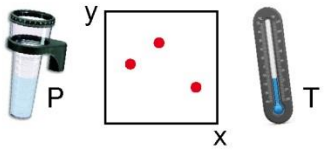
Denis Ruelland

CNRS – HydroSciences Laboratory - Montpellier, France
denis.ruelland@umontpellier.fr

How elevation affects air temperature and precipitation?

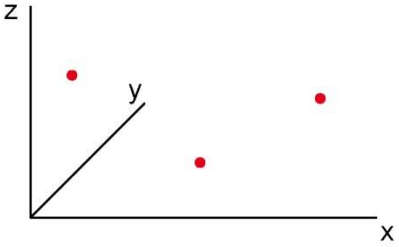


PUNCTUAL MEASURES



Errors of measures
Scattered measurements

T and P vary in space (x, y) and over time (t) and also depend on the vertical dimension (z, elevation)



In general, higher elevations cause much lower temperatures and more precipitation due to the so-called 'orographic effect' which can be expressed as elevation gradients (or lapse rates):

°C (100m)$^{-1}$
T

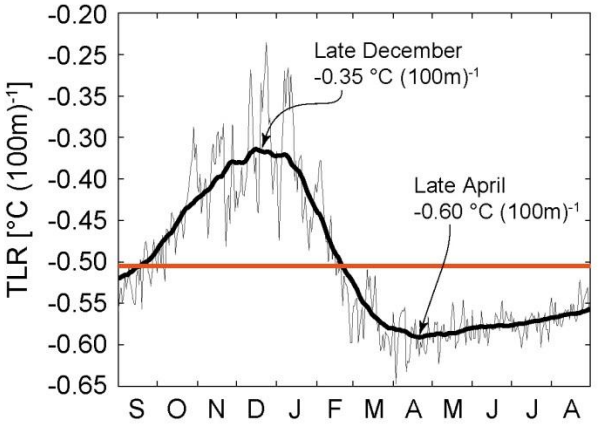


%(km)$^{-1}$
P



Temperature lapse rates (*)



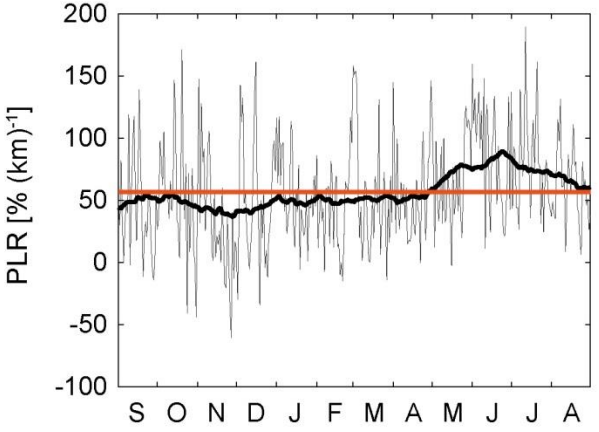


Relatively clear seasonal trend in link with solar radiation
But mean value rather around -0.65 °C (100m)$^{-1}$ (Barry and Chorley, 2010)

-0.50 °C (100m)$^{-1}$

Precipitation lapes rates (*)



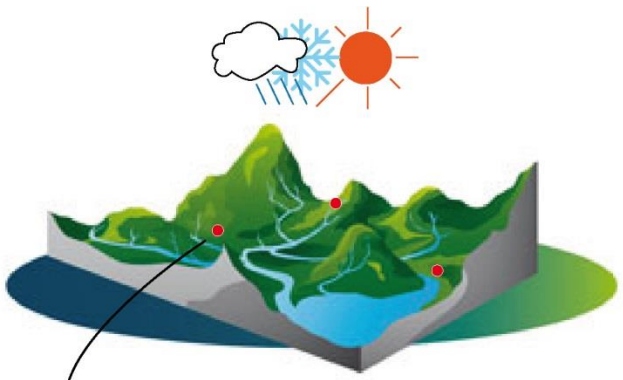


+57% (km)$^{-1}$

Extreme day-to-day variability
No clear seasonal trend
Not well known due unsufficient obs in elevation and difficulties in measuring snow

(*) Estimated from meteorological gauges (Météo-France) located between 500 and 2000 m a.s.l. in the French Alps over 2000-2016 (Ruelland, 2020)

Providing accurate meteorological inputs in moutainous regions



Schematic illustrating the uncertain 'input factors' and 'outputs' of a computer model for a given system

PUNCTUAL MEASURES

P

T

Errors of measures
Scattered measurements

SPATIALIZATION

$$Z(u) = \sum_{\alpha=1}^{n(u)} \lambda_\alpha^{ORK}(u) z(u_\alpha)$$

Interpolation model (equations)

Parameters

Spatial resolution

COMPUTER MODEL

Structures (equations)

$$y = (1+x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \cdots$$

$$y = x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Parameters

p_1 p_2 (...)

SIMULATION RESULTS

latitude

longitude

Output 1

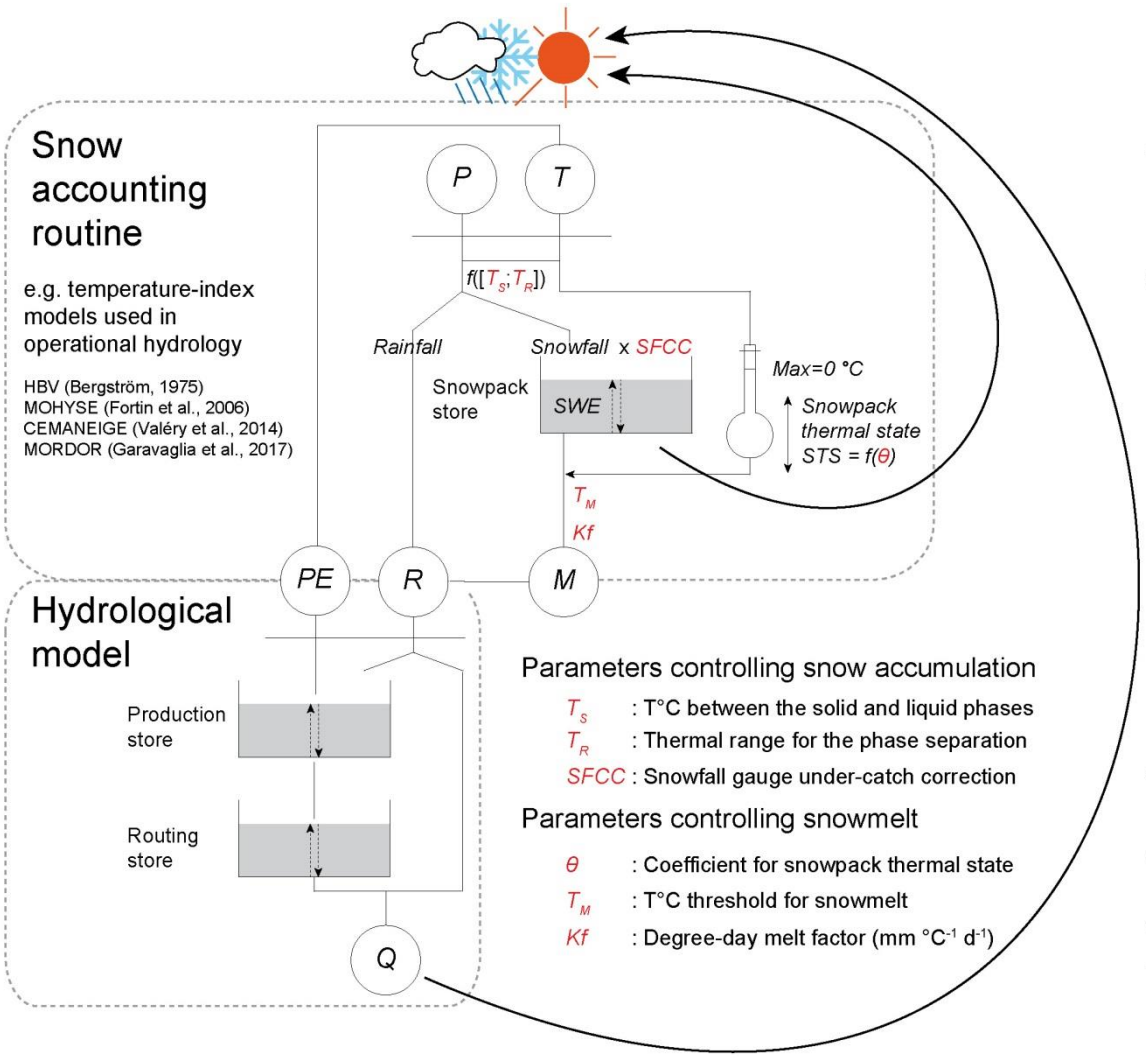
time

Output 2

time

Placing meteorological fields in a hydrological perspective

The case of conceptual snow-hydrological models



The aim of these parameters is to adapt to local snow processes, but they could be used primarily to compensate for errors in the input data without satisfactorily achieving it.

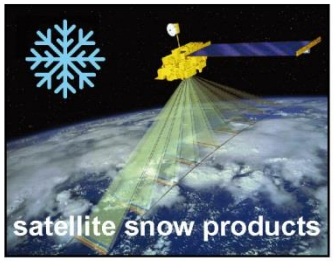


Inverting the hydrological cycle

In contrast, inverting the hydrological cycle with snow-hydrological models may help identify the dependency of the areal inputs on elevation more realistically and enable more accurate snow-hydrological simulations, while simultaneously limiting the number of free parameters

Research objectives

Can satellite snow products and streamflow measurements help assess elevation gradients of temperature and precipitation more efficiently than using the usual interpolation procedures?





To address this issue, several approaches for distributing daily temperature and precipitation are compared through sensitivity analyses of snow-hydrological models

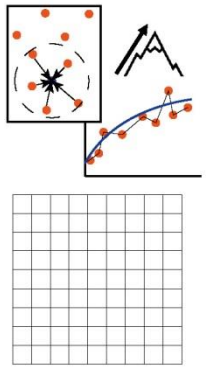


Elevation dependency in the temperature and precipitation fields is accounted for using two main strategies:

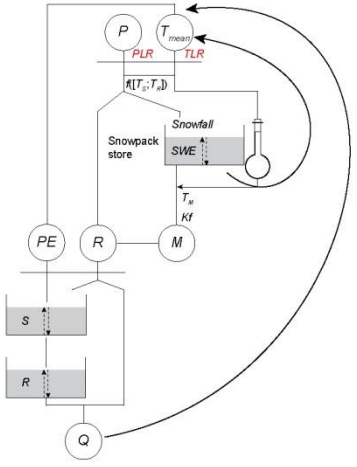
- by estimating the local and time-varying elevation gradients from the available gauge network based on **usual interpolation methods** with an external drift
- by embeding part of the distribution process into an **inverse modelling approach** via local calibrated elevation gradients correcting a first-guess distribution without elevation dependency

Usual interpolation methods

$$Z(u) = \sum_{\alpha=1}^{n(u)} \lambda_\alpha^{ORK}(u) z(u_\alpha)$$

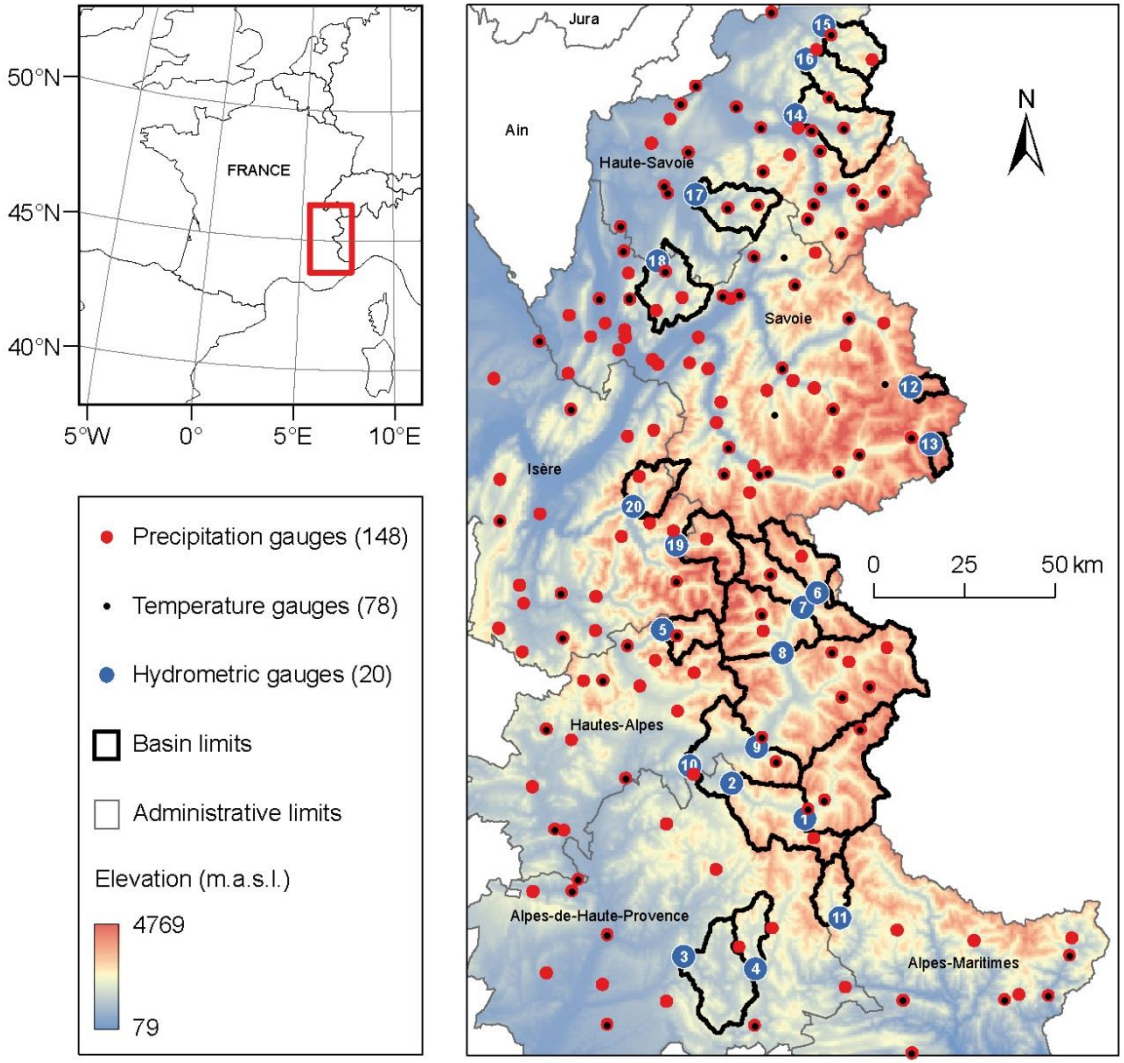


Inverse modelling approach

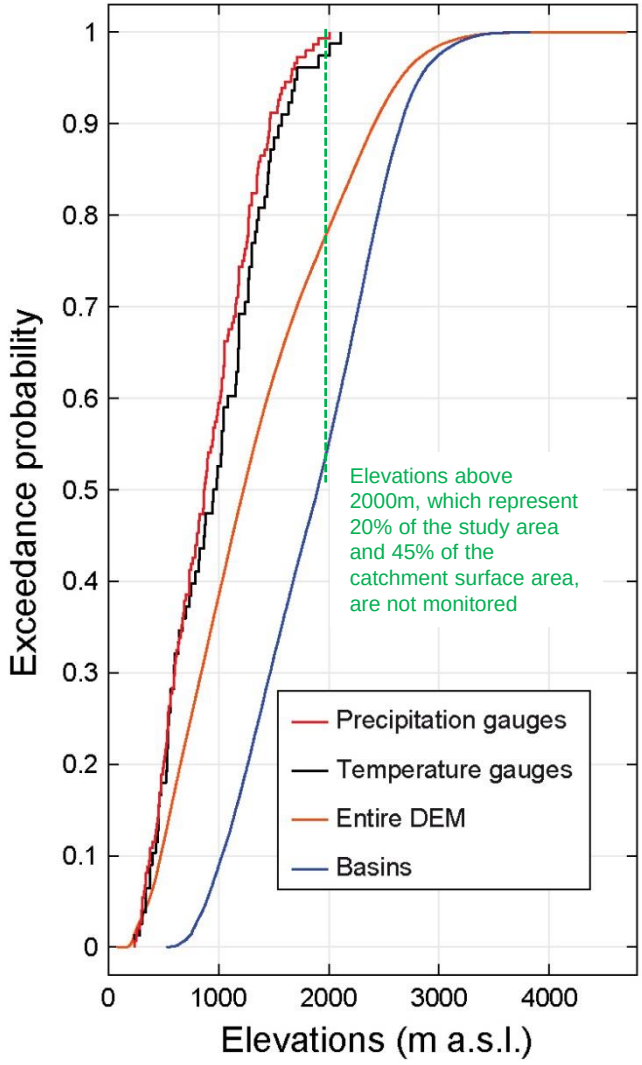


Study area and dataset: meteorological and streamflow data

Location of the P, T and Q stations



Elevation distributions

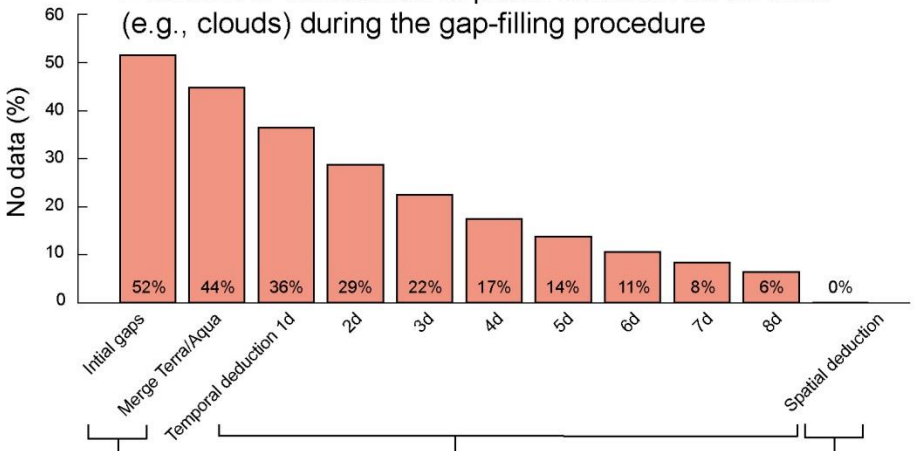


Snow cover data: gap-filling of MODIS snow products (2000–2016)

Daily MODIS snow products available from the Terra (since 2000) and Aqua (from 2002) satellites



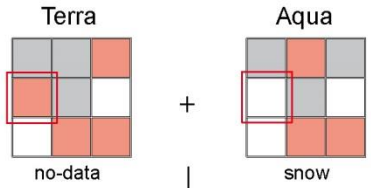
Evolution of the number of pixels classified as no-data (e.g., clouds) during the gap-filling procedure

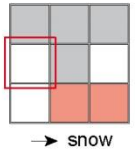


Pixels:

snow | no-snow | no-data (clouds, missing data, no decision)

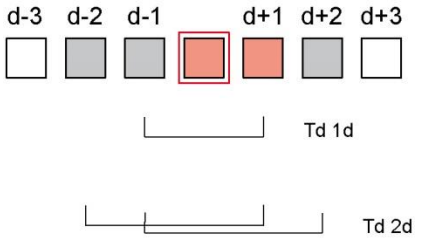
MODIS Terra/Aqua combination

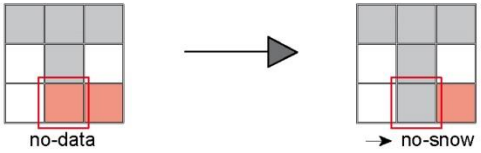




Adjacent temporal deduction (Td)

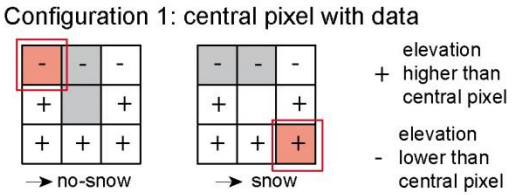
xd sliding windows

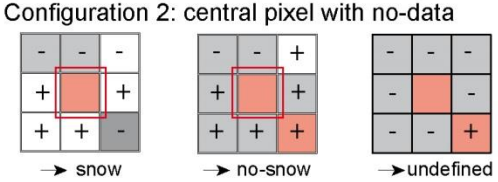




Spatial deduction (Sd)

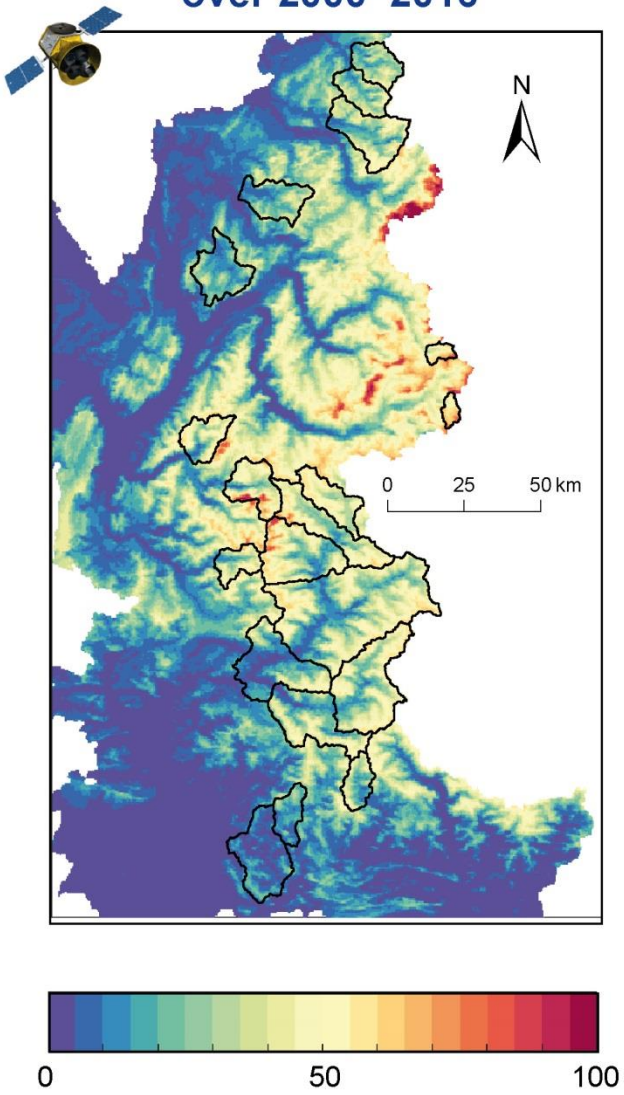
3*3 filter according to elevation and neighboring



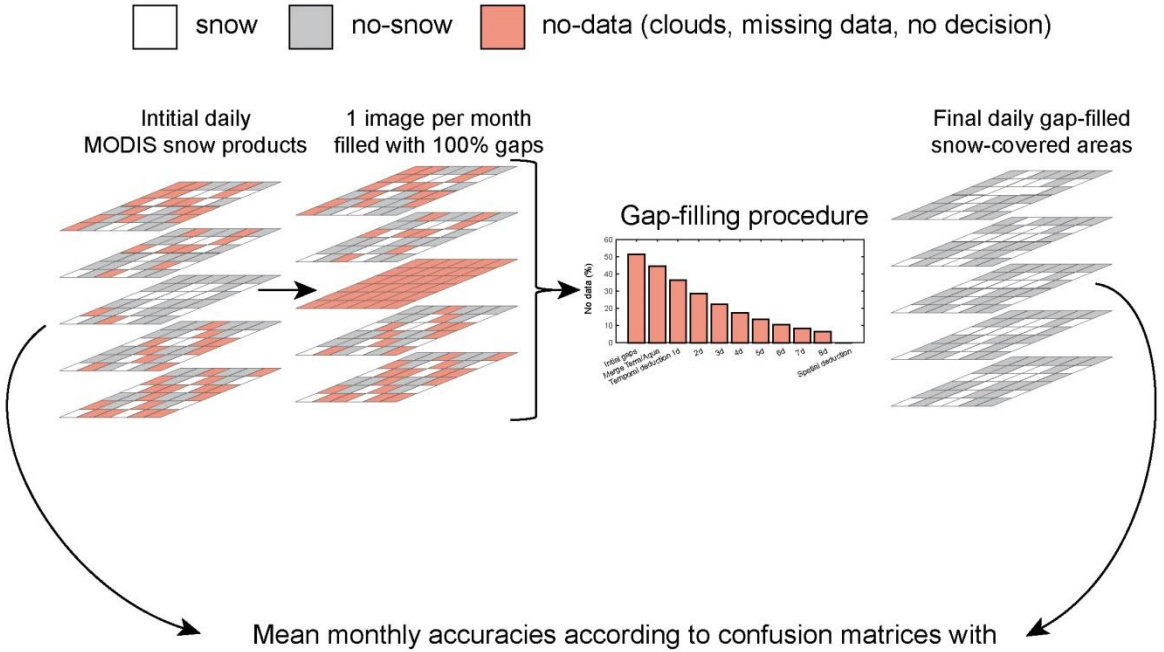


Results of gap-filling applied to MODIS snow products

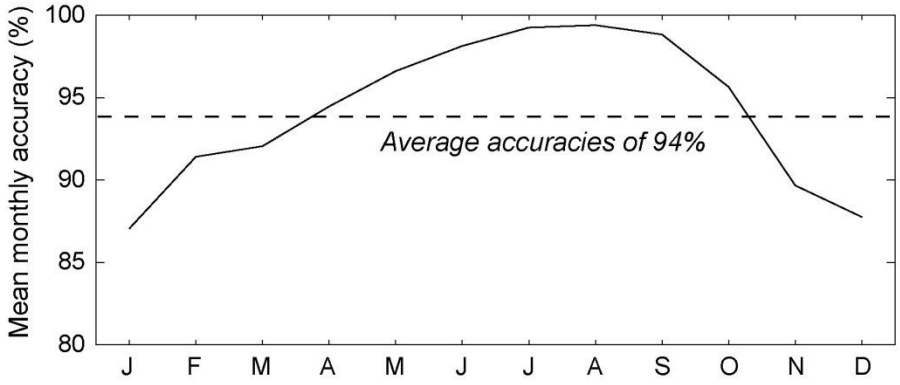
Snow cover duration (%) over 2000–2016



Validation of the gap-filling procedure



Mean monthly accuracies according to confusion matrices with one image/month, i.e. ~200 cloud-free images over 2000–2016



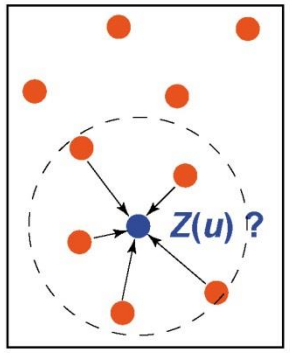
Schemes for spatialization of temperature and precipitation

Schemes for interpolating a surface from sampled points depend on:

- The method used to choose the surrounding points (neighborhood)

Closest *n* number of points

n within a given search radius



- The model used to characterise the spatial variation of the variable of interest

Deterministic approach

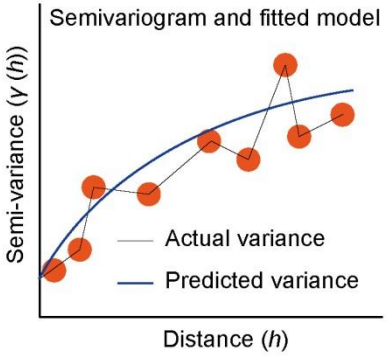
e.g. Inverse distance weighted (IDW)

$$Z_{\mathrm{IDW}}(u) = \frac{1}{\sum_{\alpha=1}^{n(u)} \lambda_\alpha(u)} \sum_{\alpha=1}^{n(u)} \lambda_\alpha(u) z(u_\alpha) \quad \text{with} \quad \lambda_\alpha(u) = \frac{1}{|u - u_\alpha|^\omega}$$

Geostatistical approach

e.g. Ordinary kriging (ORK)

$$Z_{\mathrm{ORK}}(u) = \sum_{\alpha=1}^{n(u)} \lambda_\alpha^{ORK}(u) z(u_\alpha) \quad \text{with} \quad \sum_{\alpha=1}^{n(u)} \lambda_\alpha^{ORK}(u) = 1$$



- The approach used to adjust for elevation dependency

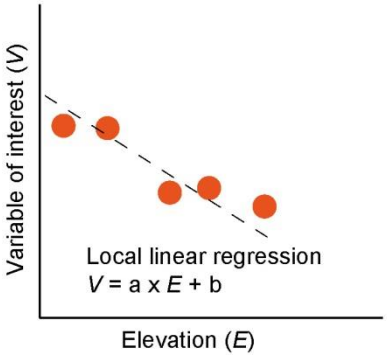
e.g. Kriging with an external drift (KED) using elevation as secondary information

$$Z_{\mathrm{KED}}(u) - \mathrm{m}(u) = \sum_{\alpha=1}^{n(u)} \lambda_\alpha^{KED}(u)[z(u_\alpha) - \mathrm{m}(u_\alpha)]$$

$$\text{with} \quad \mathrm{m}(u) = a_0(u) + a_1(u) y(u)$$



KED can account for local daily differences in topographic influence because the regression coefficients computed between the primary variable (T or P) and the secondary variable (elevation) are implicitly estimated through the kriging system within each search neighbourhood



Interpolation procedures tested

Methods

°C (100m)$^{-1}$
% (km)$^{-1}$

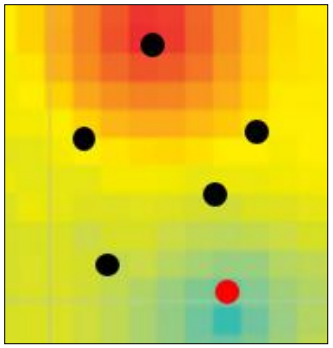


	Without elevation dependency	With elevation as external drift
Deterministic	Inverse Distance Weighted (IDW)	IDW with external drift (IED)
	$Z_{\mathrm{IDW}}(u) = \frac{1}{\sum_{\alpha=1}^{n(u)} \lambda_\alpha(u)} \sum_{\alpha=1}^{n(u)} \lambda_\alpha(u) z(u_\alpha)$	$Z_{\mathrm{IED}}(u) - m(u) = \frac{1}{\sum_{\alpha=1}^{n(u)} \lambda_\alpha(u)} \sum_{\alpha=1}^{n(u)} \lambda_\alpha(u)[z(u_\alpha) - m(u_\alpha)]$
Geostatistical	Ordinary kriging (ORK)	Kriging with external drift (KED)
	$Z_{\mathrm{ORK}}(u) = \sum_{\alpha=1}^{n(u)} \lambda_\alpha^{ORK}(u) z(u_\alpha)$	$Z_{\mathrm{KED}}(u) - m(u) = \sum_{\alpha=1}^{n(u)} \lambda_\alpha^{KED}(u)[z(u_\alpha) - m(u_\alpha)]$

Evaluation of interpolation performances

(1) Cross-validation

Leave-one-out procedure

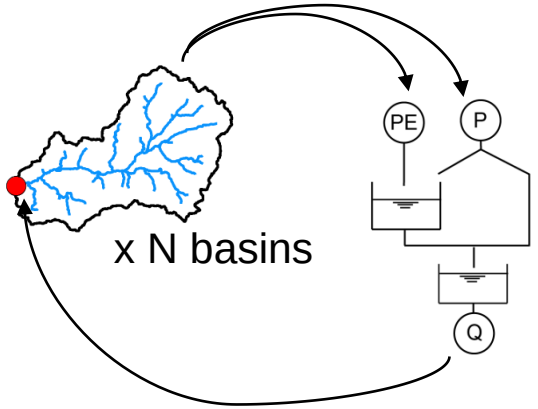


Parameter optimization
(IDW, ORK, KED and IED)

+

Efficiency of interpolations
(RMSE)

(2) Hydrological sensitivity analyses

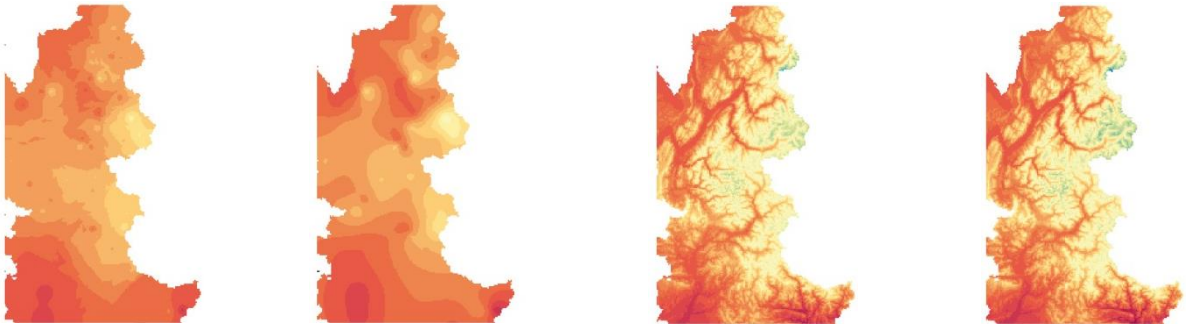


Cross-validation results and interpolation maps (2000–2016)

Temperature (78 gauges)

	Without elevation dependency		With elevation as external drift	
Methods	**IDW**	**ORK**	**KED**	**IED**
RMSE	2.16 °C	2.18 °C	1.21 °C	**1.20 °C**

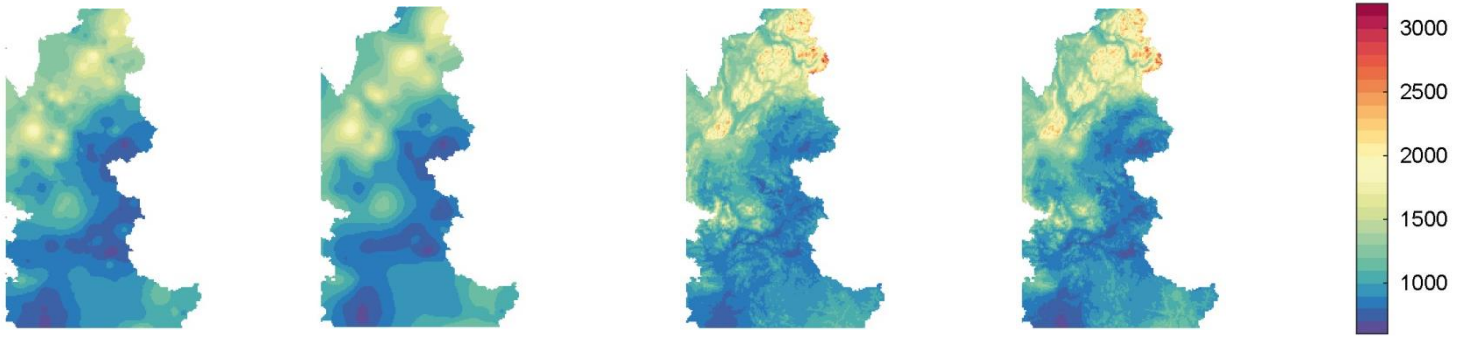
Maps



Precipitation (148 gauges)

	Without elevation dependency		With elevation as external drift	
Methods	**IDW**	**ORK**	**KED**	**IED**
RMSE	**2.83 mm**	2.86 mm	2.91 mm	2.90 mm

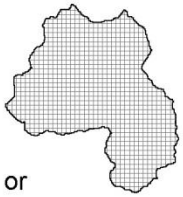
Maps



Snow accounting routine (SAR): accumulation and ablation

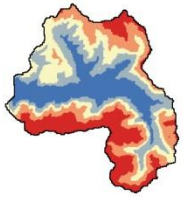
Conceptual scheme

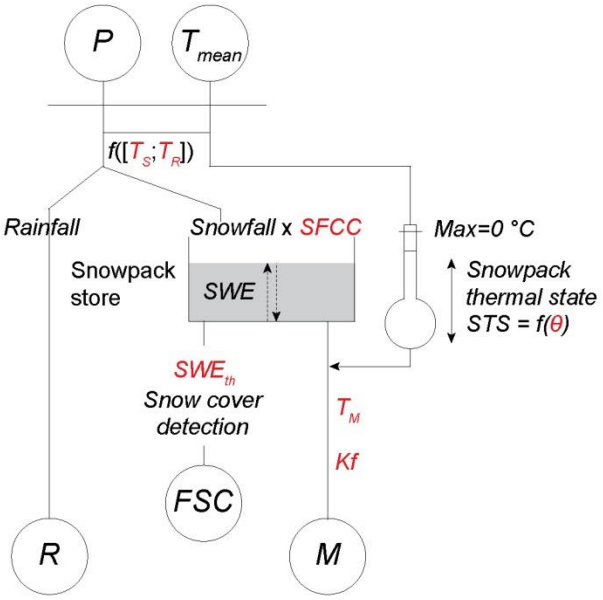
Full distribution



or

Elevation bands





Associated equations

- Snow/rain partition of total precipitation (P)

  $SolidFraction = f([T_S\ °C;\ T_R\ °C])$

  $Snowfall = P \times SolidFraction$ and $R\ (Rain) = P \times (1 - SolidFraction)$

  $Snowfall = Snowfall \times SFCC$ and $P = R + Snowfall$

- Snow accumulation and snowpack themal state (STS) updating

  $SWE = SWE + Snowfall$

  $STS = \min(\theta \times STS + (1 - \theta) \times T_{mean}, 0)$

- Snowmelt (M) computation and snowpack updating

  If $STS = 0$ and $T_{mean} > T_M$ °C, $M = \min(Kf \times (T_{mean} - T_M), SWE)$,

  Otherwise $M = 0$

  $SWE = SWE - M$

- Fractional snow-covered area (FSC) computation

  $FSC = \min(SWE / SWE_{th}, 1)$

Distributed input, output and state variables

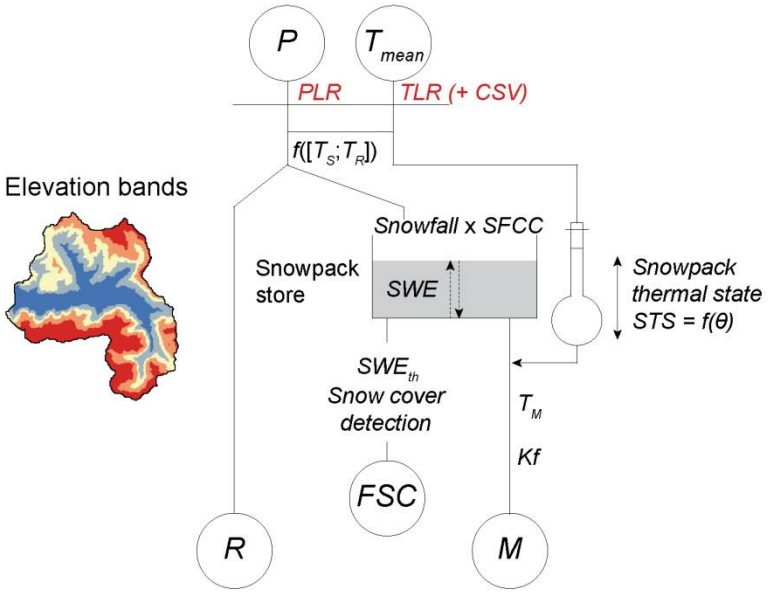
P	: Total precipitation (mm d^{-1})
T_{mean}	: Mean temperature (°C)
S	: Snowfall (mm d^{-1})
R	: Rainfall (mm d^{-1})
SWE	: Snow Water Equivalent (mm d^{-1})
M	: Melt (mm d^{-1})
FSC	: Fractional snow-covered area (% d^{-1})

Parameters

T_S	: Temperature between the solid and liquid phases (°C)
T_R	: Thermal range for the phase separation above T_S (°C)
$SFCC$	: Snowfall gauge under-catch correction factor (-)
θ	: Weighting coefficient for snowpack thermal state (-)
T_M	: Temperature threshold for snowmelt (°C)
Kf	: Degree-day melt factor (mm $°C^{-1}$ d^{-1})
SWE_{th}	: Snow water equivalent threshold to compute FSC (mm)

Introducing elevation gradients (lapse rates) as key parameters

When gridded precipitation and temperature datasets interpolated without elevation dependency are used, the SAR enables input data for each elevation band (or cell) to be modified based on two elevation gradients (*PLR* and *TLR*) used as key parameters:



Precipitation correction

$$P_i(t) = P_i^{IDW}(t) \times \left[1 + \frac{PLR}{1000} \times (y_i - y_i^{IDW}(t)\right]$$

PLR and *TLR* are the constant precipitation and temperature lapse rates to be calibrated

$T_i^{IDW}(t)$ and $P_i^{IDW}(t)$ are the mean areal temperature and precipitation interpolated based on the IDW procedure in elevation zone *i* at time step *t*

$y_i^{IDW}(t)$ is the mean areal elevation interpolated based on the IDW procedure in elevation zone *i* from the available gauges at time step *t*

y_i is the mean areal elevation extracted from the DEM in elevation zone *i*

Temperature correction

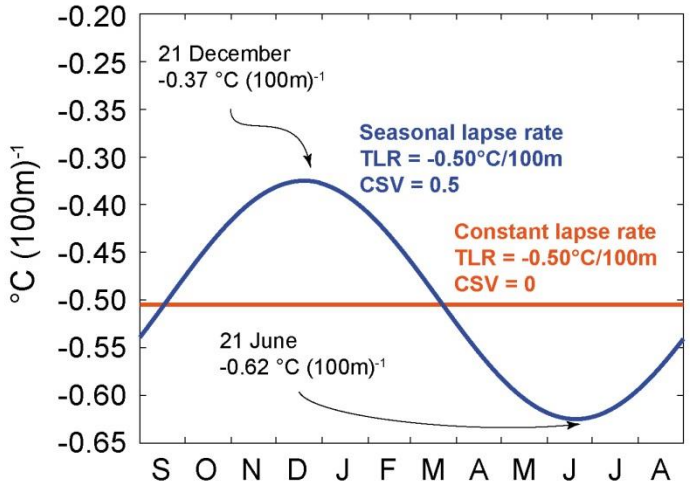
$$T_i(t) = T_i^{IDW}(t) + \left[\frac{\left(TLR + \frac{1}{2}TLR \times Si \times CSV\right)}{100} \times \left(y_i - y_i^{IDW}(t)\right)\right]$$

$$Si = \sin\left(\frac{2\pi \times (d - 80.5)}{366}\right)$$

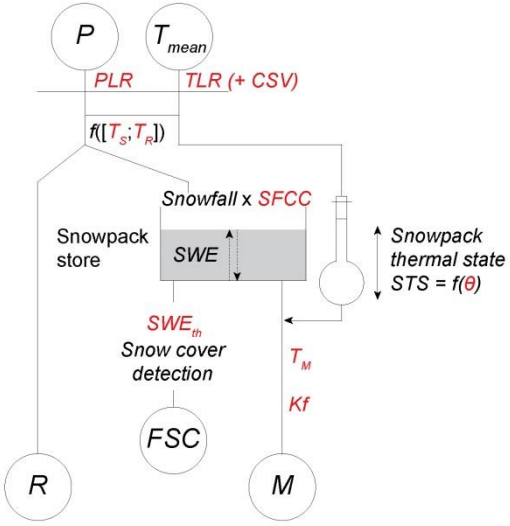
CSV is a coefficient of seasonal variation due to solar radiation to be applied to *TLR* (when set to 0, no seasonal variation is applied)

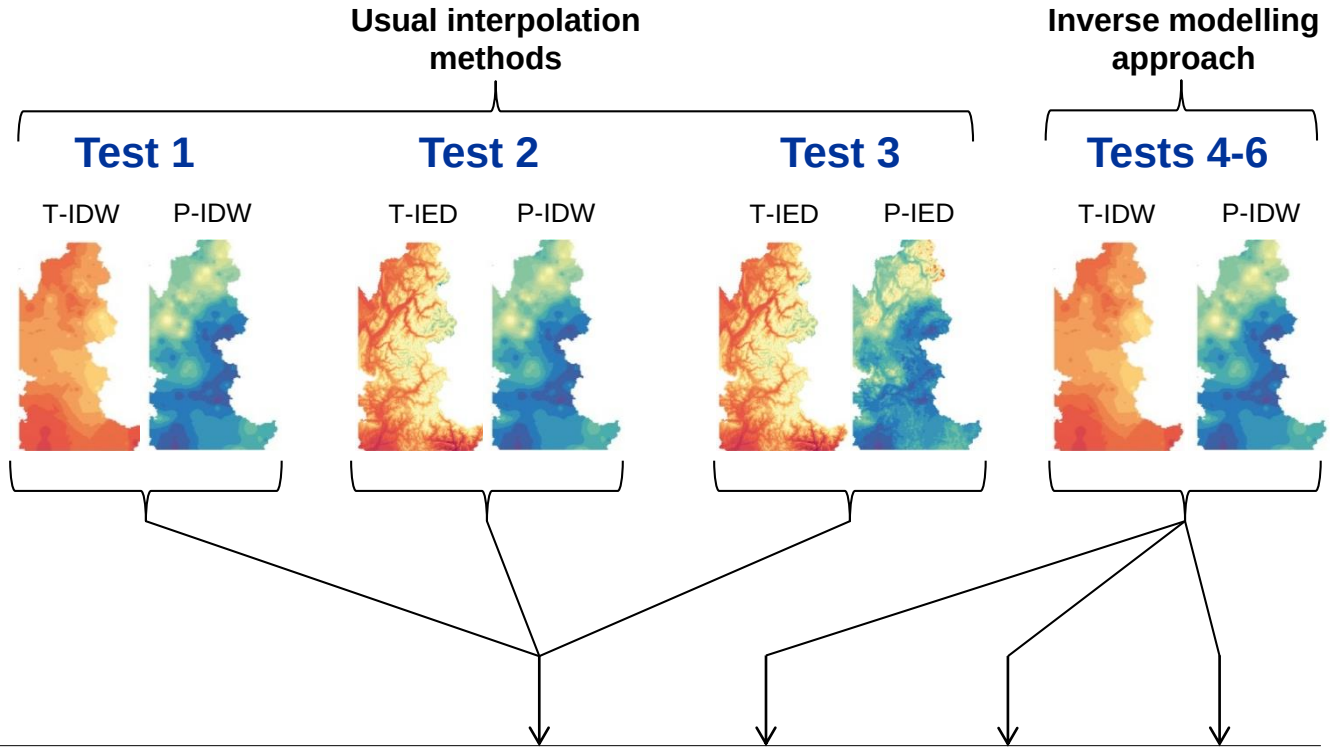
Si is an index of seasonal change in solar radiation accounting for daytime length and ranges from -1 on the 21st of December (winter solstice) to 1 on the 21st of June (summer solstice), *d* is the number of days since the 1st of January of the current year.

Examples



Six tests within the modelling experiment





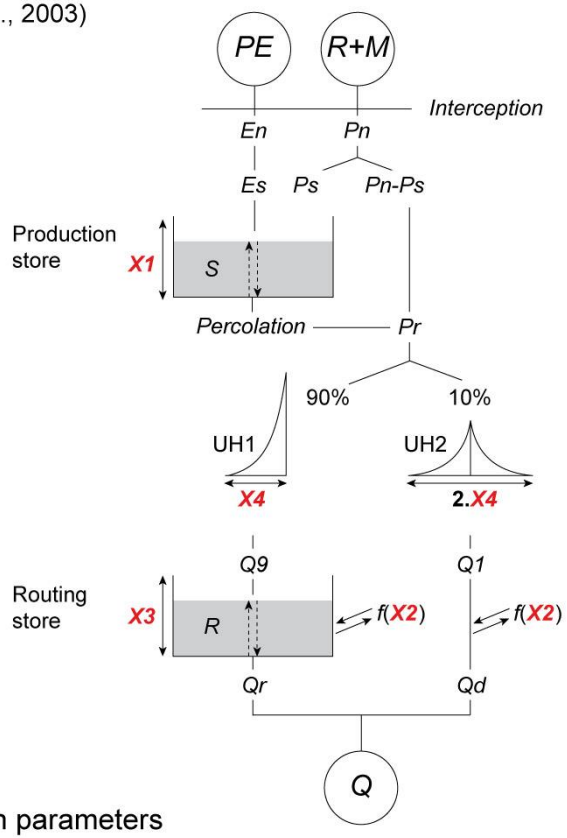
Fixed values (black) or ranges tested (red)

Param.	Meaning	Unit	Tests 1-3	Test 4	Test 5	Test 6
T_S	Temperature between the solid and liquid phase	°C	[-3; 3]	-1	-1	-1
T_R	Thermal range for the phase separation above T_S	°C	[0; 10]	4	4	4
SFCC	Snowfall gauge under-catch correction factor	-	[1; 3]	1	1	1
θ	Weighting coefficient for snowpack thermal state	-	[0; 1]	0	0	[0; 1]
T_M	Temperature threshold for snowmelt	°C	T_S + 1	0	0	0
Kf	Degree-day melt factor	$mm.°C^{-1}.d^{-1}$	[0; 10]	5	5	[0; 10]
SWE_{th}	Snow water equivalent threshold to compute *FSC*	mm	40	40	40	40
TLR	Temperature lapse rate	$°C\ (100m)^{-1}$	-	[0;-1.5]	[0;-1.5]	[0;-1.5]
CSV	Coefficient of seasonal variation applied to *TLR*	-	-	0	[0; 1]	[0; 1]
PLR	Precipitation lapse rate	$\%\ (km)^{-1}$	-	[0; 200]	[0; 200]	[0; 200]
	Number of free parameters		**5**	**2**	**3**	**5**

Two hydrological models to avoid model-specific results

GR4J

(Perrin et al., 2003)

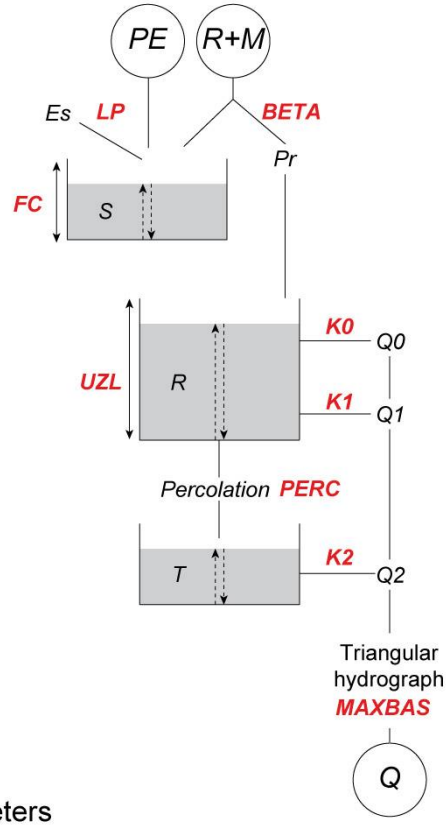


Calibration parameters

X1: Maximum capacity of the production store (mm)
X2: Inter-catchment exchange coefficient ($mm.d^{-1}$)
X3: Maximum capacity of the non-linear routing store (mm)
X4: Unit hydrograph (UH) time base (d)

HBV9

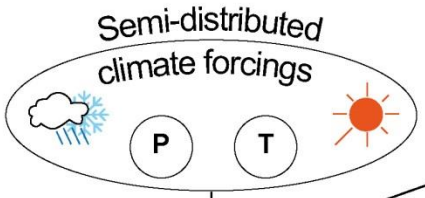
(Bergström, 1975)



Calibration parameters

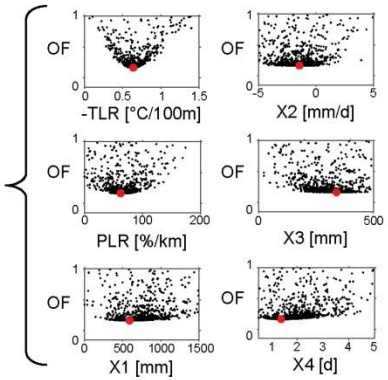
BETA : Shape coefficient of recharge function (-)
FC : Max. water storage in the unsaturated-zone store (mm)
LP : Soil moisture value above which AE reaches PE (%)
K0 : Additional recession coefficient of upper groundwater store (d^{-1})
K1 : Recession coefficient of upper groundwater store (d^{-1})
K2 : Recession coefficient of lower groundwater store (d^{-1})
PERC : Maximum percolation to lower zone ($mm.d^{-1}$)
UZL : Threshold parameter for extra outflow from upper zone (mm)
MAXBAS : Length of equilateral triangular weighting function (-)

Calibration methods: objective function and optimization algorithm



20 basins

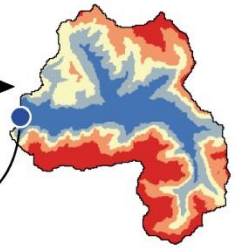
Optimization algorithm
Shuffle Complex evolution



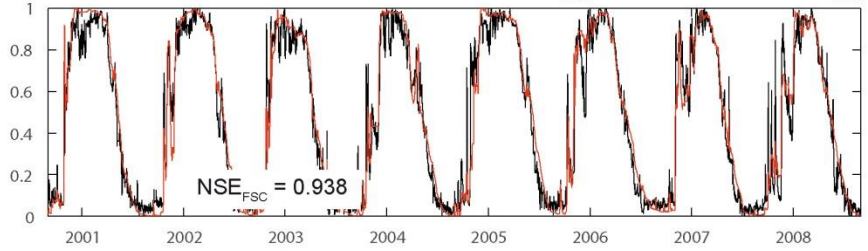
SNOW-HYDRO MODELS



Elevation bands

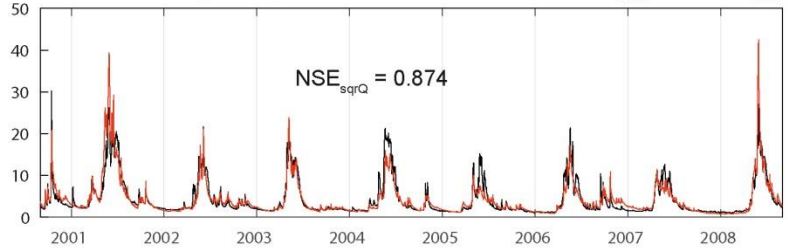


Fractional snow-cover area (FSC) — FSC_{OBS} (MODIS) — FSC_{SIM}



$$NSE_{SNOW} = \frac{1}{E}\sum_{i=1}^{E}\left(1 - \frac{\sum_{t=1}^{N}(FSC_{sim,t}^{i} - FSC_{obs,t}^{i})^2}{\sum_{t=1}^{N}(FSC_{obs,t}^{i} - \overline{FSC_{obs}^{i}})^2}\right)$$

Streamflow - Q ($m^3 s^{-1}$) — Q_{OBS} — Q_{SIM}



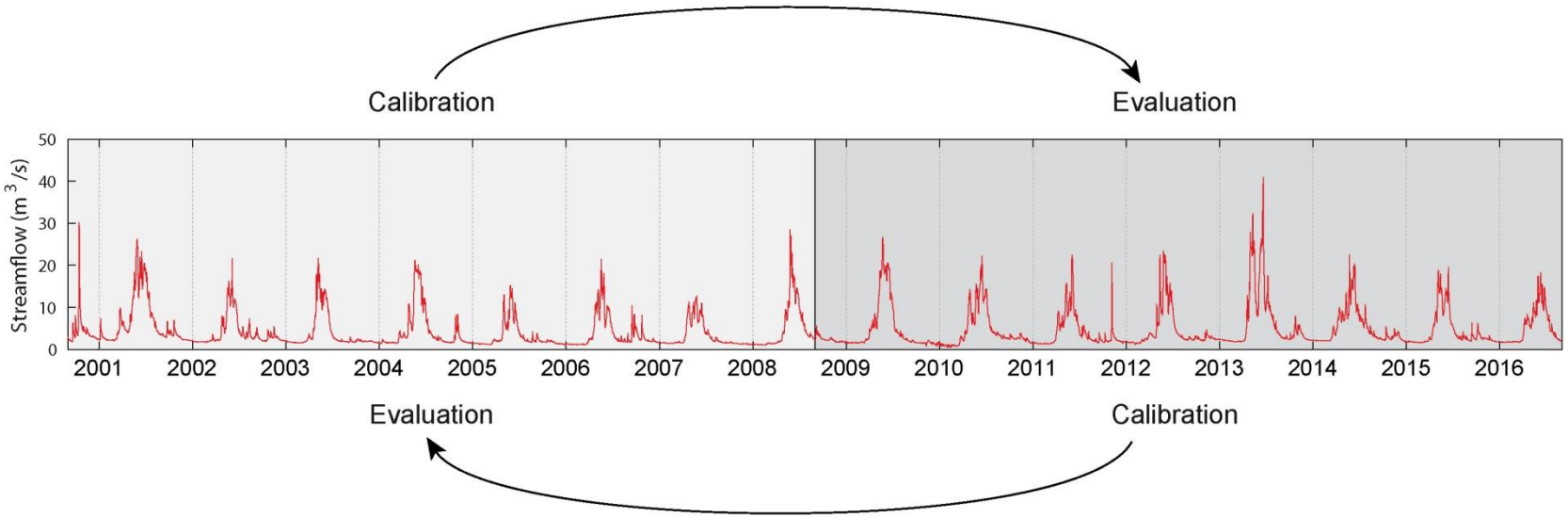
Multi-criteria composite objective function OF

$$OF = 1 - (0.5 \times NSE_{SNOW} + 0.5 \times NSE_{sqrQ})$$

$$NSE_{sqrQ} = 1 - \frac{\sum_{t=1}^{N}(\sqrt{Q_{sim,t}} - \sqrt{Q_{obs,t}})^2}{\sum_{t=1}^{N}(\sqrt{Q_{obs,t}} - \overline{\sqrt{Q_{obs}}})^2}$$

Evaluation: Split-sample tests and efficiency criteria

Split-sample tests on a 16-year periods in each basin (hydrological years)



The simulation period (2000-2016) was split into two sub-periods alternatively used for calibration and evaluation.

2 evaluations periods x 20 basins = 40 evaluation exercices

Four efficiency criteria in evaluation

Snow dynamics

$$\mathrm{NSE_{SNOW}} = \frac{1}{E}\sum_{i=1}^{E}\left(1 - \frac{\sum_{t=1}^{N}(FSC_{sim,t}^{i} - FSC_{obs,t}^{i})^{2}}{\sum_{t=1}^{N}(FSC_{obs,t}^{i} - \overline{FSC_{obs}^{i}})^{2}}\right)$$

High flows

$$\mathrm{NSE_{Q}} = 1 - \frac{\sum_{t=1}^{N}(Q_{sim,t} - Q_{obs,t})^{2}}{\sum_{t=1}^{N}(Q_{obs,t} - \overline{Q_{obs}})^{2}}$$

Low flows

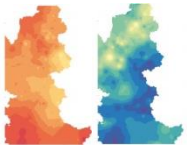
$$\mathrm{NSE_{lnQ}} = 1 - \frac{\sum_{t=1}^{N}(\log Q_{sim,t} - \log Q_{obs,t})^{2}}{\sum_{t=1}^{N}(logQ_{obs,t} - \overline{logQ_{obs}})^{2}}$$

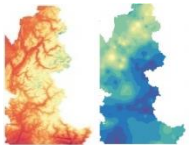
Volume error

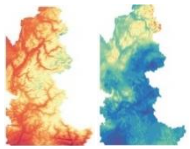
$$\mathrm{VE_{C}} = 1 - \frac{\left|\sum_{t=1}^{N} Q_{sim,t} - \sum_{i=1}^{N} Q_{obs,t}\right|}{\sum_{t=1}^{N} Q_{obs,t}}$$

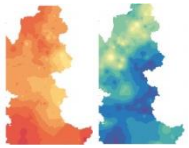
Sensitivity of the snow-hydro simulations to the elevation gradients

Test #	T input	P input	Calibrated SAR parameters (excluding hydro. models)
1	T-IDW	P-IDW	T_S, T_R, *SFCC*, *θ*, *Kf*
2	T-IED	P-IDW	T_S, T_R, *SFCC*, *θ*, *Kf*
3	T-IED	P-IED	T_S, T_R, *SFCC*, *θ*, *Kf*
4	T-IDW	P-IDW	*TLR*, *PLR*
5			*TLR*, *CSV*, *PLR*
6			*TLR*, *CSV*, *PLR*, *θ*, *Kf*

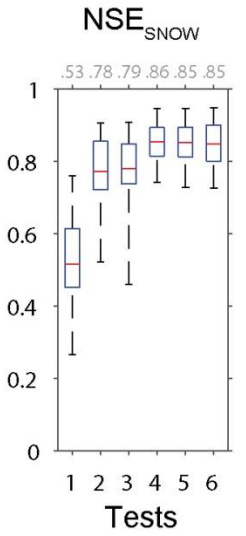


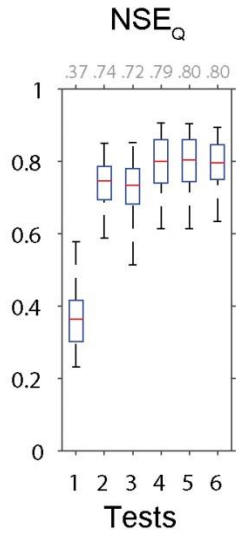


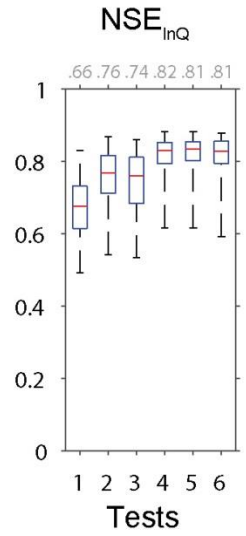


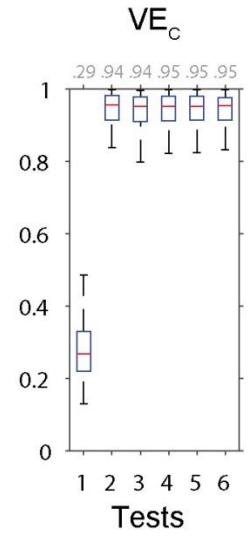


SAR + GR4J

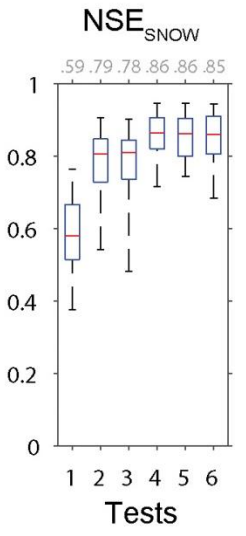


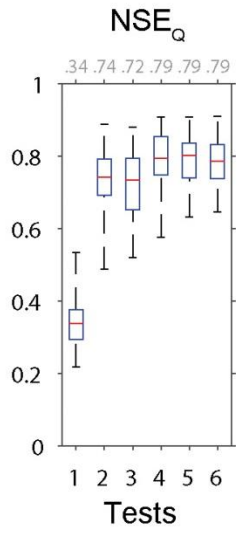


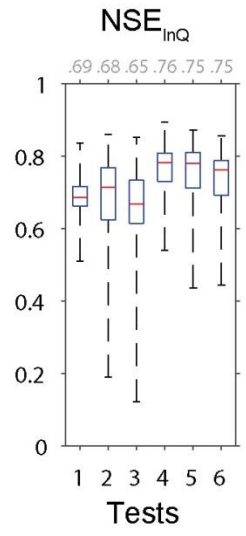


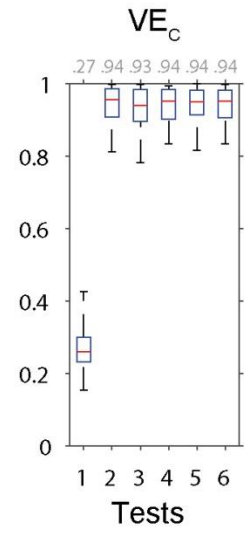


SAR + HBV9



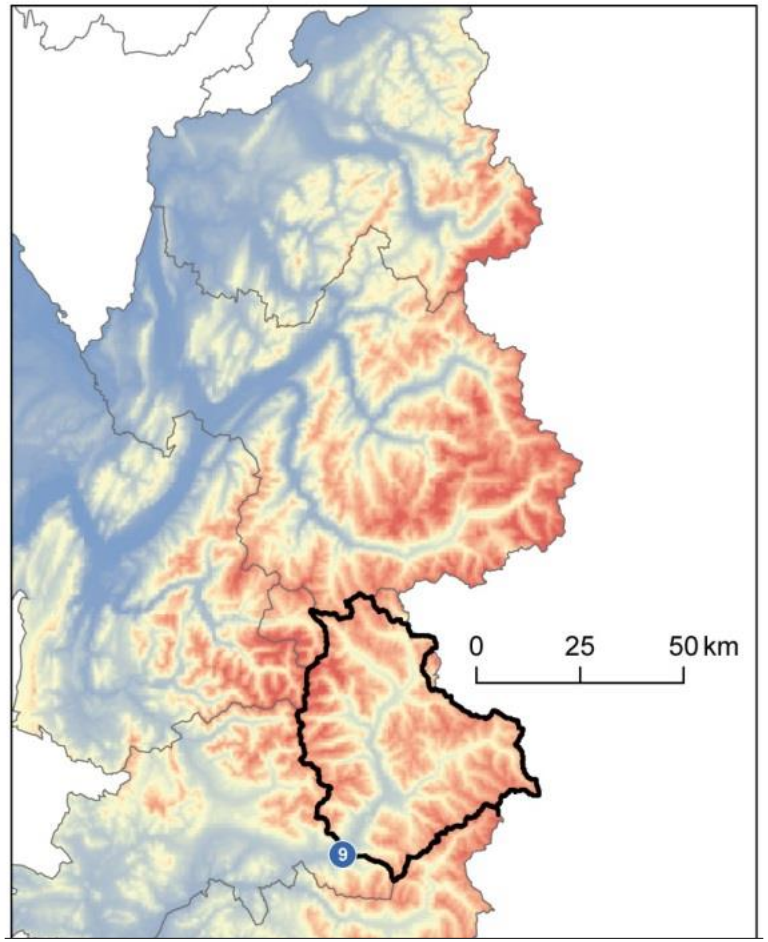




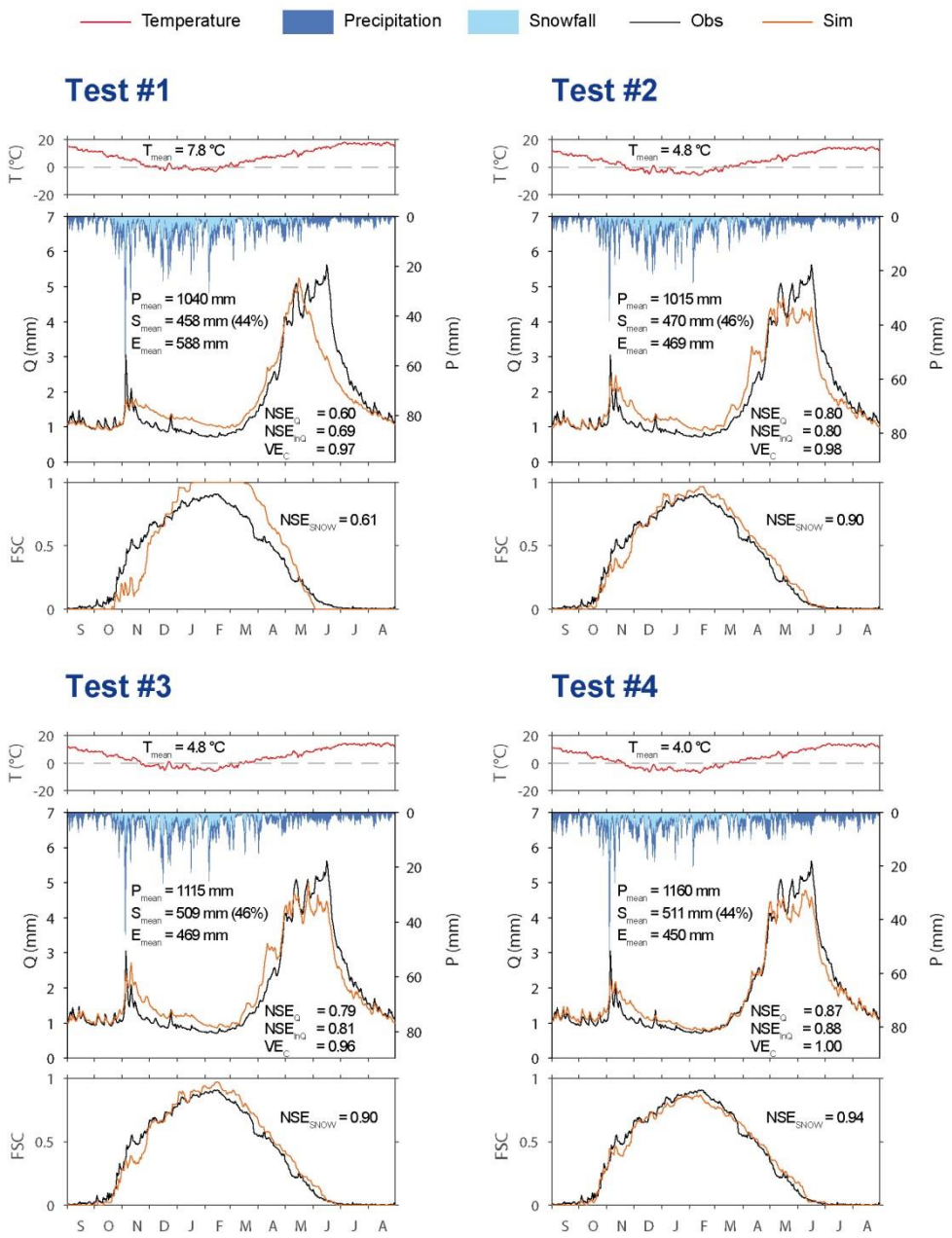


Comparison of the snow-hydrological simulations (example)

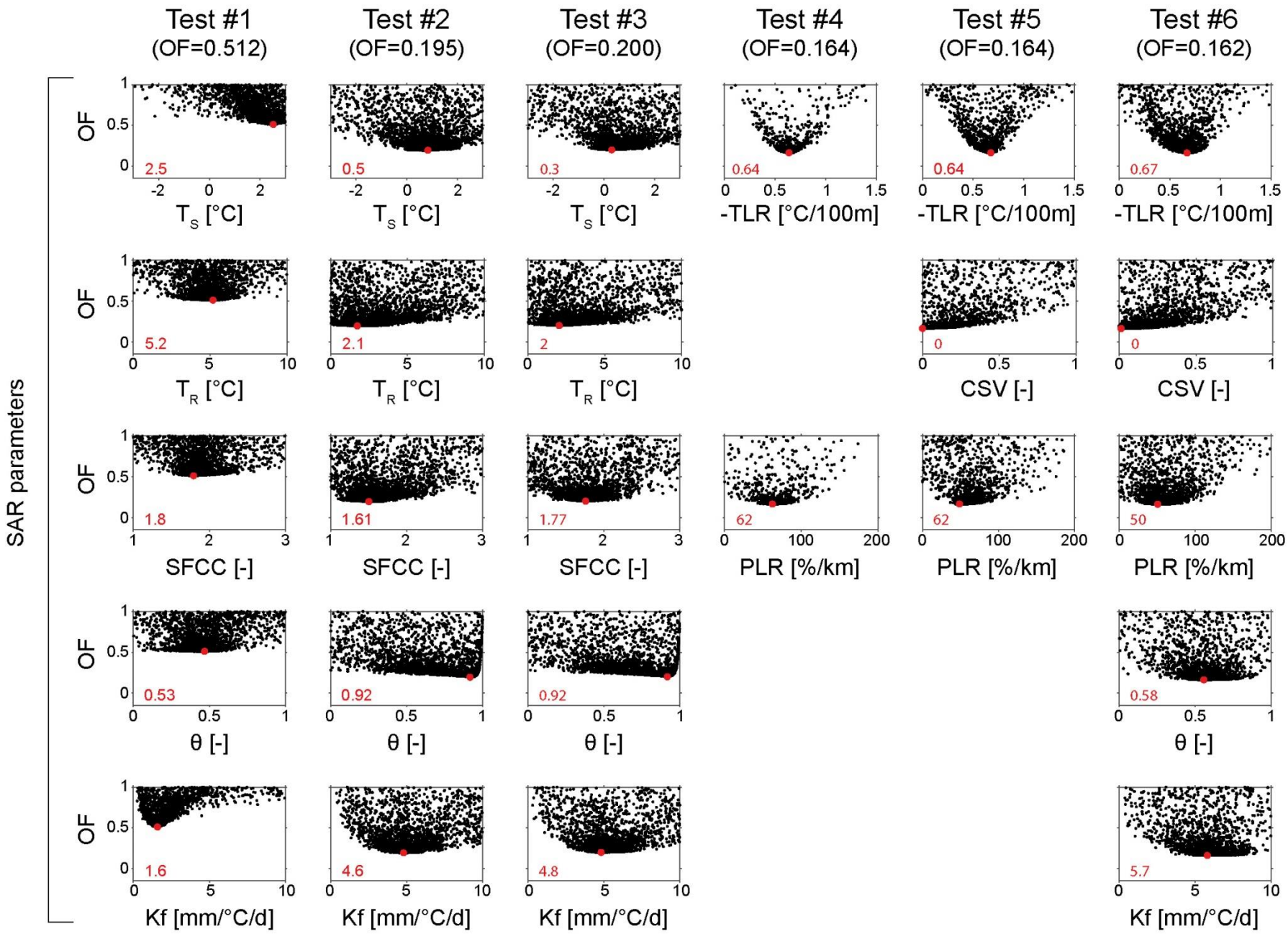
Durance at Serre-Ponçon (3580 km²)



Test #	T input	P input	Calibrated SAR parameters (excluding hydrological models)
1	T-IDW	P-IDW	T_S, T_R, *SFCC*, *θ*, *Kf*
2	T-IED	P-IDW	T_S, T_R, *SFCC*, *θ*, *Kf*
3	T-IED	P-IED	T_S, T_R, *SFCC*, *θ*, *Kf*
4	T-IDW	P-IDW	*TLR*, *PLR*

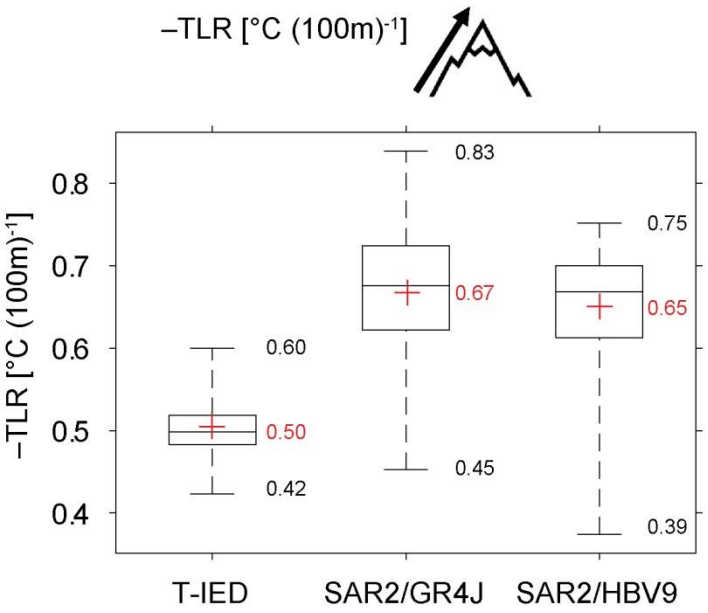


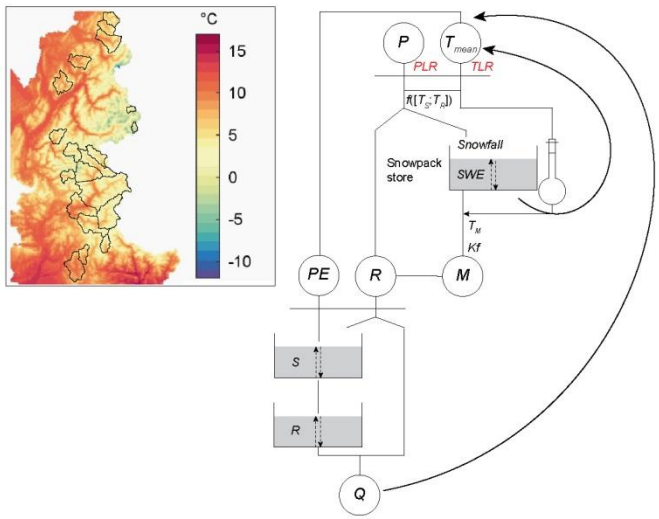
Identifiability of the SAR parameters during calibration (example)



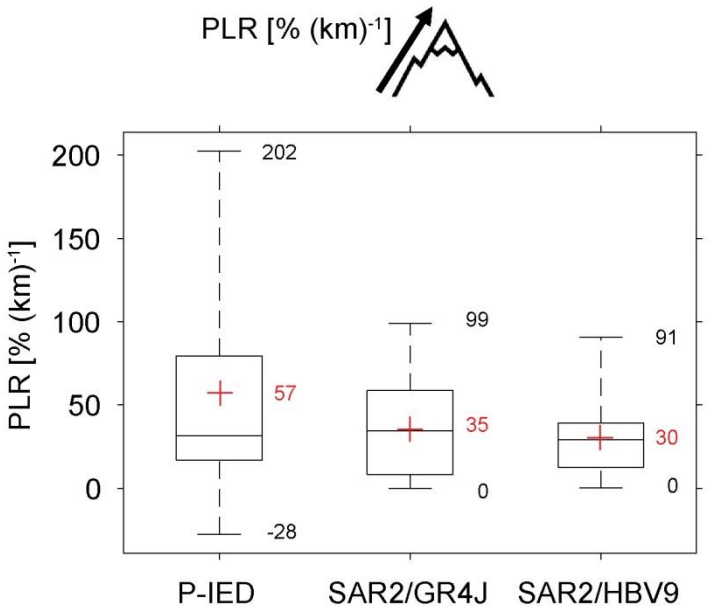
Ranges of the elevation gradients depending on the method

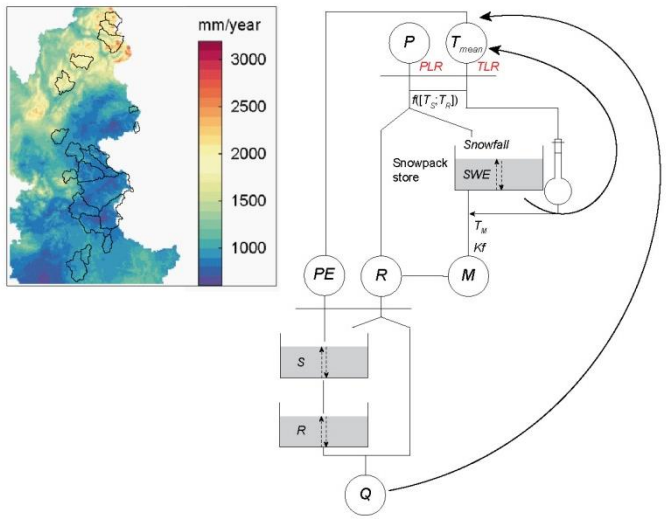
Temperature lapse rates





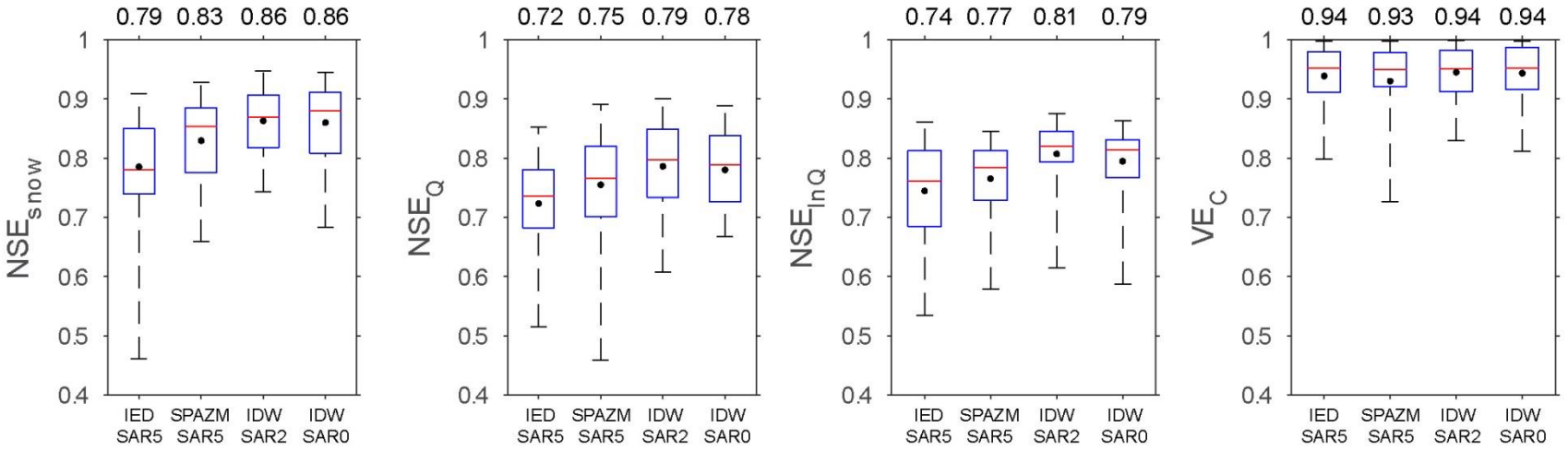
Precipitation lapes rates



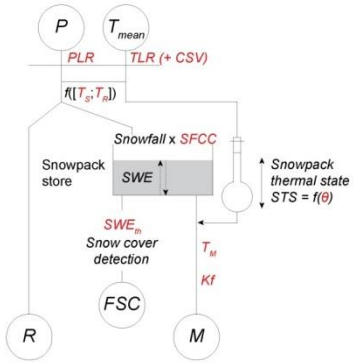


Comparison with the EDF SPAZM database

SAR/GR4J



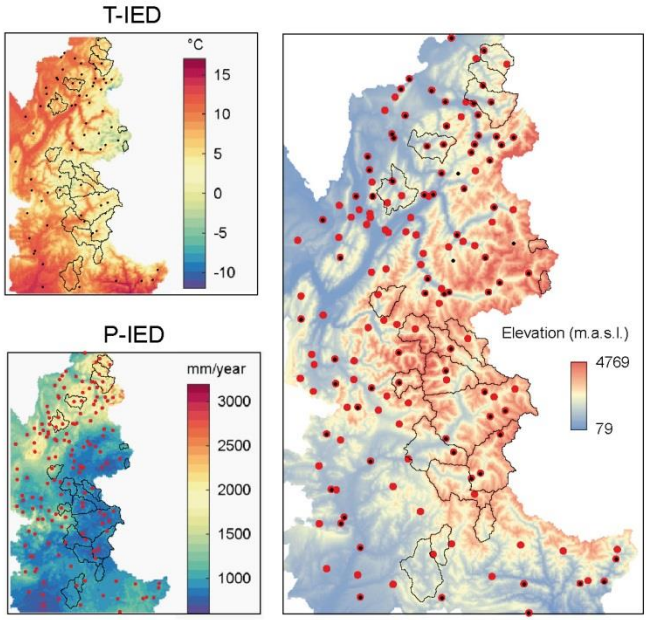
Param.	Meaning	Unit	Fixed values (black) or ranges tested (red) SAR5	SAR2	SAR0
T_S	Temperature between the solid and liquid phase	°C	[-3; 3]	-1	-1
T_R	Thermal range for the phase separation above T_S	°C	[0; 10]	4	4
SFCC	Snowfall gauge under-catch correction factor	-	[1; 3]	1	1
θ	Weighting coefficient for snowpack thermal state	-	[0; 1]	0	0
T_M	Temperature threshold for snowmelt	°C	T_S + 1	0	0
Kf	Degree-day melt factor	mm.°C^{-1}.d^{-1}	[0; 10]	5	5
SWE_{th}	Snow water equivalent threshold to compute *FSC*	mm	40	40	40
TLR	Temperature lapse rate	°C (100m)$^{-1}$	-	[0;-1.5]	**-0.67**
CSV	Coefficient of seasonal variation applied to *TLR*	-	-	0	0
PLR	Precipitation lapse rate	% (km)$^{-1}$	-	[0; 200]	**35**
	Number of free parameters		**5**	**2**	**0**

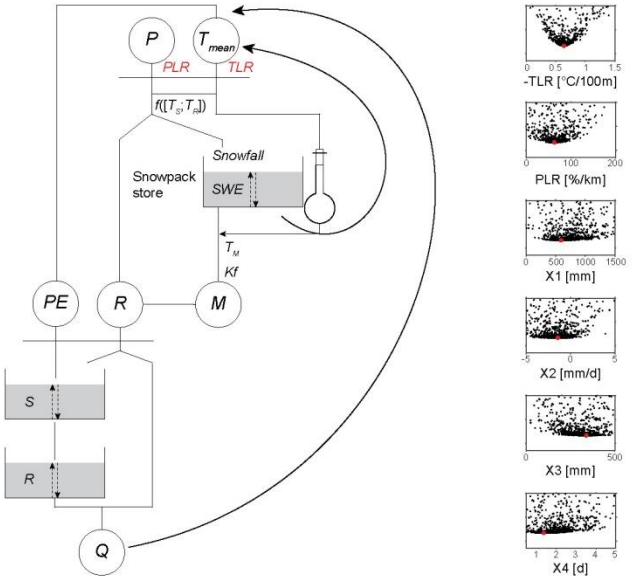


Concluding remarks

Main findings

- Interpolation methods accounting for elevation dependency from scattered gauges result in inaccurate inputs for snow-hydrological models
- The lack of high-elevation stations seriously limits correct estimation of lapse rates of temperature and precipitation, which, in turn, affects the performance of the snow-hydrological simulations due to imprecise estimates of temperature and precipitation volumes.
- Instead, calibrating constant local elevation gradients of T and P using an inverse snow-hydrological modelling approach significantly improves the simulation of snow cover and streamflow dynamics.
- Allowing additional parameters to be calibrated in the SAR brings very limited improvements, showing that the T and P lapse rates play a first-order role for the snow-hydrological simulations and should be considered as key parameters through more parsimonious parametrisation.





Thank you for listening. Any questions?



For more details, please refer to:

Ruelland, D. (2020). Should altitudinal gradients of temperature and precipitation inputs be inferred from key parameters in snow-hydrological models? Hydrol. & Earth Syst. Sci., 24, 2609–2632.